Supplement File of Probabilistic End-to-End Vehicle Navigation in Complex Dynamic Environments with Multimodal Sensor Fusion

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Fig. S1: Three levels of traffic density in our *DeepTest* benchmark.

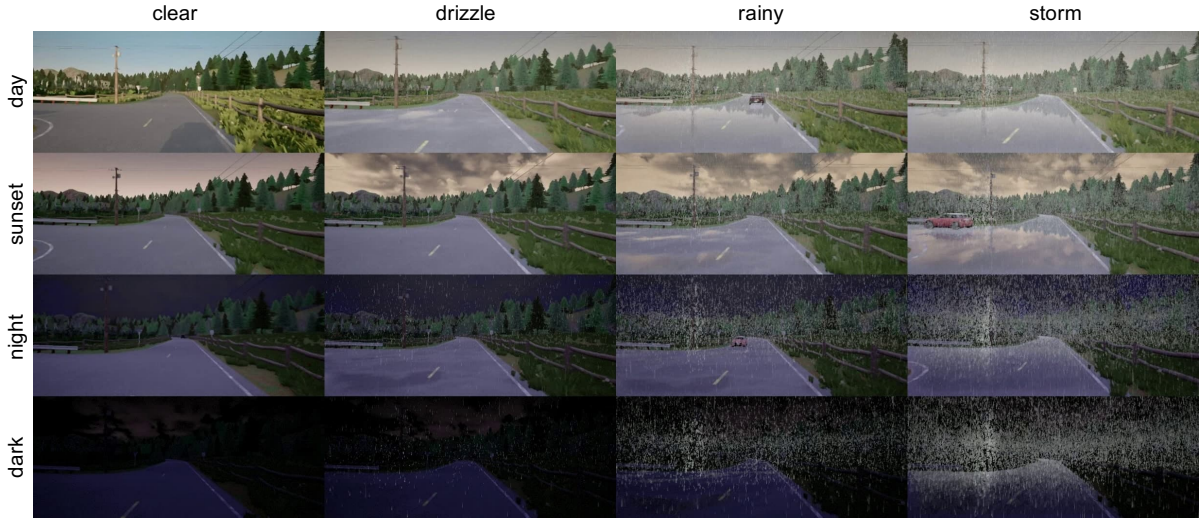

Fig. S2: Varied illumination (*clear, sunset, night* and *dark*) and weather (*clear, drizzle, rainy* and *storm*) conditions considered in this work.

A. Traffic Densities

We set three levels of traffic density for our *DeepTest* benchmark: *empty*, *regular* and *dense*. The specific settings are shown in Table. S1 and can be viewed in Fig. S1.

TABLE S1: Different traffic densities in this work

Type	Number of pedestrians	Number of vehicles
Empty	0	0
Regular	40~75	60
Dense	60~150	80~120

B. Illumination and Weather Conditions

This paper involves four illumination conditions (i.e., *daytime*, *sunset*, *night* and *dark*) and four weather conditions (i.e., *clear*, *drizzle*, *rainy* and *storm*). The specific settings can be viewed in Fig. S2.

C. PID Control

To translate the planned motion distribution into action $\boldsymbol{a}_2$, two PID controllers are designed for lateral (steering) and longitudinal (throttle and brake) control. The parameters are shown in Table S2.

TABLE S2: PID parameters used in this work

Type	Proportional (P)	Integral (I)	Derivative (D)
Lateral	0.70	0.00	0.00
Longitudinal	0.25	0.20	0.00